Chandrasekar Vasan

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EDUCATION

Master of Science in Computer Science, The University of Texas at Arlington | Arlington, TX, USA **Jan 2025 - Dec 2026**
Coursework: Artificial Intelligence | Design & Analysis of Algorithms | Data Analysis & Modelling Techniques ***(GPA: 3.9/4.00)***
Bachelor of Technology in Information Technology, Anna University | Chennai, TN, India **Aug 2020 - Jun 2024**
Coursework: Machine Learning | Operating Systems | Cloud Computing | Software Engineering ***(GPA: 4.00/4.00)***

WORK EXPERIENCE

The University of Texas at Arlington **Aug 2025 - Present**
Graduate Research Assistant – System Administrator **Arlington, Texas, USA**

- Improved diagnostic success in offline replays from 62% with retrieval-only assistance to 84% across 40 held-out incidents, five runs each (+22 pp; 95% CI +12 to +32.5 pp)
- Designed and evaluated a tool-selection heuristic; on the same held-out replays, averaged 5.6 versus 8.2 calls per investigation and $0.31 versus $0.42 per successful investigation compared with fixed ordering
- Ran development-set ablations: retrieval added 10 pp success versus no retrieval; critique reduced incorrect tool calls from 9% to 4% at 8% additional cost; memory added 10% cost without success gains and was removed
- Fixed a crash between tool execution and checkpoint persistence using durable tool receipts and idempotency keys, improving recovery from 39/40 to 40/40 interrupted workflows with zero duplicate remediation requests on failure-injection reruns

eNoah iSolution **Jan 2024 - Apr 2024**
Software Engineer **Chennai, Tamil Nadu, India**

- Prototyped an isotonic-calibrated XGBoost loan model, reducing expected decision cost by 18% (95% CI 6.9% to 28.4%) versus Logistic Regression on 1,000 held-out synthetic applications under assumed 4:1 false-approval/rejection penalties
- Piloted a stateful CRM assistant with four volunteer reviewers across 80 synthetic cases; timed, counterbalanced tests reduced median review time from 8 to 6 minutes (25%), with 95% agreement with predefined eligibility-rule references in both conditions
- Integrated Dockerized Python APIs with React, CRM, and PostgreSQL/MongoDB; all 80 synthetic cases completed the intake, scoring, and review workflow with persisted model outputs, reviewer decisions, and overrides

Kun Auto Co Pvt Ltd **Jan 2023 - Dec 2023**
Software Engineer **Chennai, Tamil Nadu, India**

- Built a React, Node.js, and PostgreSQL vehicle-valuation prototype, passing 50 synthetic workflow tests covering input validation, Python model inference, price retrieval, and persisted reviewer adjustments
- Implemented GitHub Actions pipelines with Docker-based API and database checks, failing builds on prediction-contract or persistence-test errors and producing reproducible model-service images
- Benchmarked gradient boosting against ridge regression on 200 held-out synthetic profiles generated from depreciation rules and noise, observing 15% lower MAE within that benchmark (95% paired-bootstrap CI 13.0% to 17.0% reduction)

PROJECTS

DisruptionOS | Agentic Logistics Recovery

- Built a stateful logistics-recovery agent that retrieved fleet conditions, investigated missing information, and generated approval-ready recovery plans for 50-stop delivery schedules under vehicle breakdowns and changing deadlines
- Improved simulated on-time delivery from 82% to 90% (+8 pp; 95% paired-scenario CI: +3 to +13 pp) versus a fixed workflow using the same optimizer across 200 held-out scenarios, holding fleet capacity and overtime budgets constant
- Implemented versioned approvals, idempotent dispatch actions, and durable execution state, blocking stale recommendations and preventing retried tool calls from duplicating approved dispatch changes

ExperimentPilot | Budget-Aware Autonomous Experimentation

- Engineered an experimentation agent combining Bayesian optimization, isolated trial execution, and persistent experiment state to select configurations, investigate failed runs, and make stopping decisions under compute and quality constraints
- Reduced total experimentation cost by 24% versus standard cost-aware Bayesian optimization across 30 held-out workloads, achieving predefined quality–latency targets on 27/30 workloads in both conditions; included failed trials and LLM inference costs
- Detected 12/12 seeded measurement faults involving warm-up bias, unit errors, and cache contamination through controlled reruns and metamorphic checks before accepting AI-generated experiment runners

CausalLift | Incremental Marketing Impact Modeling

- Engineered an experimentation agent combining Bayesian optimization, isolated trial execution, and persistent experiment state to select configurations, investigate failed runs, and make stopping decisions under compute and quality constraints
- Reduced total experimentation cost by 24% versus standard cost-aware Bayesian optimization across 30 held-out workloads, achieving predefined quality–latency targets on 27/30 workloads in both conditions; included failed trials and LLM inference costs
- Detected 12/12 seeded measurement faults involving warm-up bias, unit errors, and cache contamination through controlled reruns and metamorphic checks before accepting AI-generated experiment runners

TECHNICAL SKILLS

- **Languages & Data:** Python, SQL, JavaScript, NumPy, pandas, PostgreSQL, MongoDB
- **AI & Machine Learning:** scikit-learn, XGBoost, LightGBM, LangGraph, LLM Tool Calling, Structured Outputs, Stateful Agent Workflows
- **Decision Science:** Causal Inference, Uplift Modeling, Contextual Bandits, EconML, Vowpal Wabbit, OR-Tools, SimPy
- **Evaluation & Experimentation:** Calibration, Ablation Studies, Bootstrap CIs, Off-Policy Evaluation (IPS, Doubly Robust)
- **Software Engineering:** FastAPI, React, Node.js, Docker, Linux, Git, GitHub Actions, pytest, REST APIs

CERTIFICATIONS & ACHIEVEMENTS

- **Certifications:** AWS Cloud Practitioner | Google IT Support | IBM Linux & Shell | Google Data Analytics | CKA (in progress)
- **Achievement:** Second place, TECHNOPHILIA 2022 symposium, for an SVM-based Parkinson's detection model (87% accuracy)